4 Simple square demonstration

We now apply the construction of Section 2 and the protocol of Section 3 to the simplest case we can check by hand: a square plate pulled uniformly at one end.

The plate occupies $0 \le x, y \le L$ with $L = 1$ m, thickness $t = 0.01$ m, $E = 200$ GPa and $\nu = 0.3$, in plane stress (Fig. 1a). The left edge is clamped, $u_x = u_y = 0$, and the right edge carries the uniform traction $\bar{\tau} = (p, 0)$ with $p = 100$ MPa, a total force of $ptL = 1$ MN. The top and bottom edges are free. The mesh has $m \times m$ square cells, each cut along one diagonal, giving $(m+1)^2$ nodes and $2m^2$ triangles. The clamp fixes $2(m+1)$ displacement components, leaving $N = 2m(m+1)$ unknowns. A free bar under this load would stretch by $pL/E = 0.5$ mm, so we expect the right edge to move by about this amount.

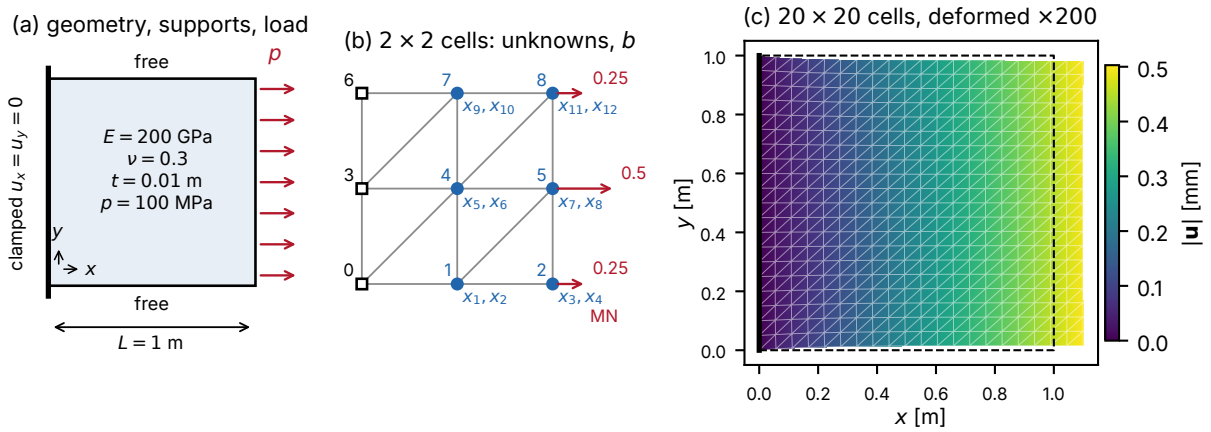


Figure 1: (a) Geometry, supports and load. (b) The 2×2-cell mesh: clamped nodes (squares), free nodes (dots) with their unknowns $x_1, \ldots, x_{12}$, and the nodal forces forming b. (c) Original outline (dashed) and deformed 20×20-cell plate, displacements amplified by 200; colour is the displacement magnitude in mm.

4.1 A small system solved directly

For $m = 2$ there are 9 nodes and 8 triangles. Clamping nodes 0, 3 and 6 leaves $N = 12$ unknowns, two per free node (Fig. 1b). As the prescribed displacements are zero, $b = f_F$. Each half of the right edge carries 0.5 MN, shared equally by its two end nodes, so the right-edge nodes receive 0.25, 0.5 and 0.25 MN in the x-direction and all other entries of b are zero.

The 12×12 matrix A is exactly symmetric, and 86 of its 144 entries are nonzero. The zeros couple nodes that share no triangle, such as nodes 1 and 7. Every entry is an integer multiple of $Et/(40(1-\nu^2)) \approx 5.49 \times 10^7$ N/m; the full matrix is in the repository file `draft/experiments/results/square/small_A_integer.csv`. Its eigenvalues range from 1.07×10^8 to 1.03×10^{10} N/m, so $\kappa_2(A) = 95.8$ and the system is well conditioned.

Pivoted $PA = LU$, the explicit inverse and Cholesky all give relative residuals η (Eq. (3.4)) below 10^{-15}, at rounding level. The right-edge nodes move $u_x = 0.511$, 0.482 and 0.468 mm from bottom to top, already close to 0.5 mm on this coarse mesh.

4.2 A larger system and the iterative methods

For $m = 20$ the mesh has 441 nodes, 800 triangles and $N = 840$ unknowns. Now $\kappa_2(A) = 9104$, and the scaled matrix S of Eq. (3.2) has largest eigenvalue 2.96, so $\rho_J = 1.96$. Table 1 and Fig. 2 compare all methods on this system.

The simple iterations behave as Section 3 predicts. Since $\rho_J = 1.96 > 1$, one error component of Jacobi nearly doubles and changes sign each sweep, so the method diverges and is stopped

Table 1: All methods on the 840-unknown system (Section 3 protocol, best of three times). An asterisk marks a method that did not reach $\eta < 10^{-8}$ within 20,000 iterations.

Method	Iterations	Final η	Error vs LU	Time (ms)
Sparse LU	–	3.5×10^{-14}	–	2.3
Jacobi*	49	1.3×10^{10}	3.4×10^{7}	0.65
Damped Jacobi ($\omega = 0.5$)*	20,000	4.8×10^{-6}	8.7×10^{-5}	284
Gauss–Seidel	19,136	9.996×10^{-9}	1.7×10^{-7}	7,180
Steepest descent*	20,000	3.1×10^{-6}	4.2×10^{-5}	541
CG	242	8.8×10^{-9}	9.6×10^{-10}	5.4
Diagonal PCG	226	8.8×10^{-9}	4.9×10^{-10}	6.8
MINRES	236	9.4×10^{-9}	5.9×10^{-9}	13.7
Full GMRES	228	9.5×10^{-9}	6.0×10^{-9}	186

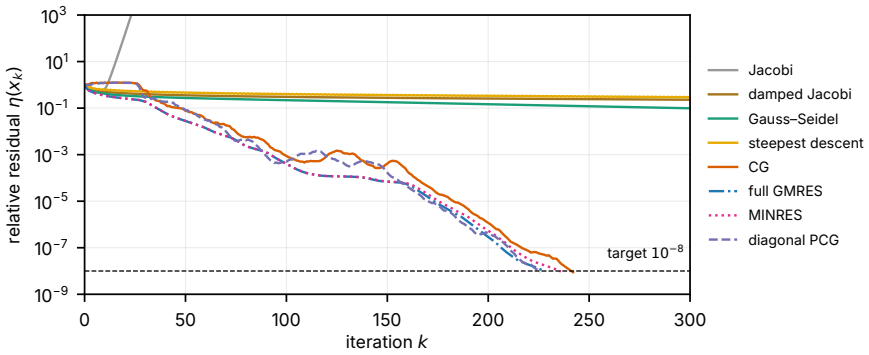


Figure 2: Relative residual $\eta(x_k)$ over the first 300 iterations on the 840-unknown system, with the target 10^{-8} dashed. The Krylov methods reach it in 226–242 iterations; the other methods have barely started.

after 49 sweeps. Damped Jacobi and steepest descent converge, but too slowly to reach the target within 20,000 iterations. Gauss–Seidel reaches it only after 19,136 sweeps and 7.2 s.

CG needs 242 iterations. Bound (3.3) with $\kappa = 9104$ guarantees the target within about $\frac{1}{2}\sqrt{\kappa}\,\ln(2 \times 10^{8}) \approx 912$ iterations, so CG does about four times better than its worst case [2]. Diagonal preconditioning saves only 16 iterations, because the diagonal of A is nearly constant on this uniform mesh.

Full GMRES uses about as many iterations as CG but takes 35 times longer. It ignores the symmetry of A and must store and orthogonalize against the whole Krylov basis, so each step costs more than the last [1]. At this small size, sparse LU (2.3 ms) is faster than every iterative method.

4.3 Deformation and validation

Figure 1c shows the 20×20 solution. The right edge moves between 0.492 and 0.497 mm, close to the expected 0.5 mm. Near the loaded end the top and bottom edges move inwards by about 0.08 mm each, matching the free lateral contraction $\nu pL/E = 0.15$ mm of the whole width. The clamp prevents this contraction at $x = 0$, which gives the tapered shape.

The solution passes four checks. All 42 prescribed components are exactly zero. The reactions of Eq. (2.4) balance the applied force of 10^{6} N and moment of 5×10^{5} N m to within 5×10^{-9} N and N m. The five methods that met the target agree with LU to a relative error of at most 1.7×10^{-7}. Finally, the patch test of Section 3 on the same mesh reproduces the exact linear field to a relative error of 2.6×10^{-14}, which confirms assembly, loading and reduction. The clamped plate has no exact solution; Section 6 estimates its discretization error.

References

[1] Yousef Saad, *Iterative methods for sparse linear systems*, second ed., SIAM, Philadelphia, 2003.

[2] Jonathan Richard Shewchuk, *An introduction to the conjugate gradient method without the agonizing pain*, Technical Report CMU-CS-94-125, School of Computer Science, Carnegie Mellon University, 1994, Edition $1\frac{1}{4}$.